

Uncertainty modelling in AI and engineering

Lecture 1: Uncertainty, evidence theory

Thierry Denœux

Université de technologie de Compiègne, France
<https://www.hds.utc.fr/~tdenoeux>

ISCC, Fall 2026

Part I

Uncertainty

Outline

- 1 What is uncertainty?
- 2 Classical models
- 3 Beyond classical models

Uncertainty is everywhere

Dennis Lindley's description

There are things we know to be true, things we know to be false, and many things whose truth or falsity is not known to us.

Uncertainty may concern:

- the **past**: who was the author of a text?
- the **future**: what will next year's economic growth be?
- the **present**: what is the value of an unknown physical constant?

Uncertainty is a consequence of lack of knowledge.

Uncertainty is a matter of degree

We usually have **some** information, but not enough to identify the true answer with certainty.

Climate sensitivity example (IPCC report)

The likely range of equilibrium climate sensitivity has been narrowed (as compared to the previous report) to 2.5°C to 4.0°C (with a best estimate of 3.0°C) based on multiple lines of evidence, including improved understanding of cloud feedbacks.

- This statement suggests two complementary notions:
 - ▶ what is sufficiently **supported** by evidence (“likely”)
 - ▶ what is **compatible** with evidence (“possible”)
- Uncertainty can decrease, and our beliefs can change, when new evidence is obtained and combined with a previous body of evidence.

Aleatory vs. epistemic uncertainty

Aleatory uncertainty

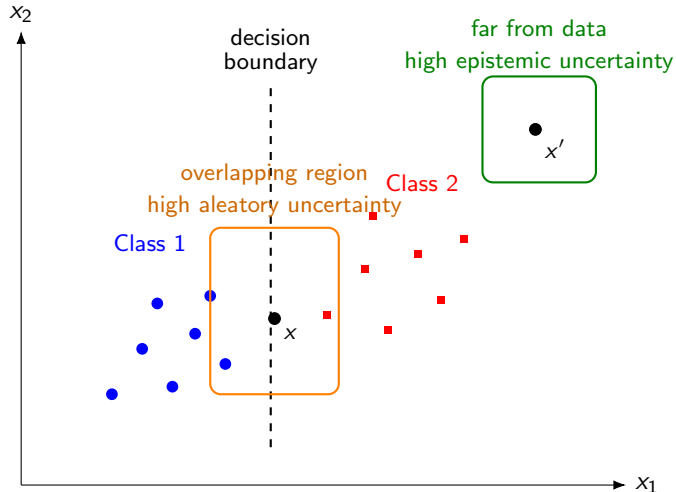
Uncertainty due to **variability** in the data-generating process, such as the result of tossing a coin.

Epistemic uncertainty

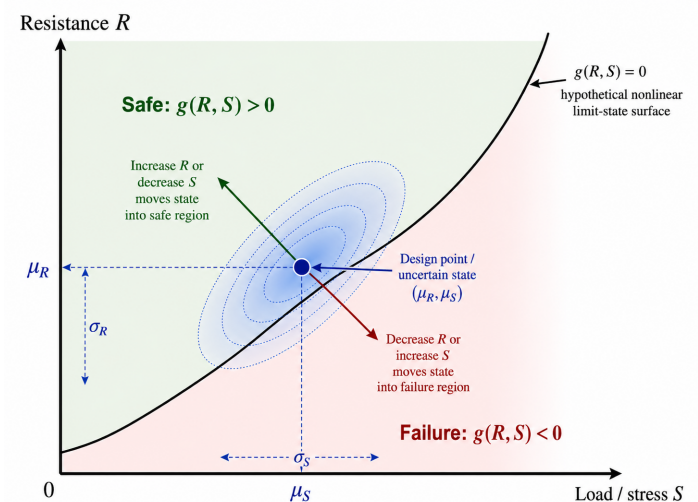
Uncertainty due to **lack of knowledge**, such as uncertainty about a unique future event or an unknown physical quantity.

In statistics, machine learning and engineering, both sources of uncertainty often coexist.

Uncertainty in machine learning



Uncertainty in reliability analysis



A basic formal setting

We consider a question $\mathcal{Q}$ with one and only one correct answer.

Unknown and frame

- X : unknown answer to the question;
- Θ : set of all possible answers, called the **frame of discernment**;
- evidence provides only **partial information** about X .

The central problem of the course is:

How can partial information be represented, propagated and combined?

Outline

- 1 What is uncertainty?
- 2 Classical models**
- 3 Beyond classical models

Set-based representations

The simplest representation of partial information is a subset $A \subseteq \Theta$.

Interpretation

$$X \in A.$$

The true value is only known to belong to a set of possible values.

Interval analysis

This idea underlies **interval analysis**: if

$$x \in [-1, 1], \quad y \in [1, 2],$$

then for $z = \exp(x + y^2)$, **interval arithmetic** gives

$$y^2 \in [1, 4], \quad x + y^2 \in [0, 5], \quad z \in [1, e^5].$$

Strengths and limits of sets

Strength

If the inputs are guaranteed to lie in given sets, and if the model is correct, the output set is **guaranteed** to contain the true output.

Limitation

Sets do not express degrees of doubt: a value is either possible or impossible.

Consequences:

- results may be too conservative to be useful;
- or too confident because some uncertainty has been ignored;
- high-dimensional boxes (Cartesian products of intervals) quickly become impractical.

Probability theory

- Probability theory is the most widely used formalism for uncertainty.
- A **probability measure** requires assigning precise numerical probabilities to all relevant events.
- This is natural for repeatable random phenomena, i.e., aleatory uncertainty.
- It is questionable when information is scarce, qualitative, vague, or based on expert judgement, i.e., epistemic uncertainty.
- How can we represent evidence telling us that $X \in [a, b]$, and nothing more, using a probability distribution?

Laplace's principle of indifference (PI)

If there is no reason to regard one possibility as more likely than another, then all possibilities should be assigned equal probability.

Paradoxes

Continuous case

- If we only know that $X \in [a, b]$, $\text{PI} \Rightarrow X \sim \text{Unif}([a, b])$.
- But if $Y = 1/X$, then Y is not uniformly distributed on $[1/b, 1/a]$.
- The result depends on the chosen parametrisation, which is often arbitrary.

Discrete case

- Assume that $A = \{\theta_1, \theta_2\}$. According to PI, $X \in A$ translates to

$$p(\theta_1) = p(\theta_2) = 1/2.$$

- If θ_2 is split into two possible values θ_{21} and θ_{22} , we get

$$p(\theta_1) = p(\theta_{21}) = p(\theta_{22}) = 1/3.$$

- Why should the probability of θ_1 depend on the degree of detail with which θ_2 is described?

Outline

- 1 What is uncertainty?
- 2 Classical models
- 3 Beyond classical models

Evidence theory

Evidence theory generalises both the set-based representations and probability theory by allowing probability mass to be assigned to **subsets** of Θ .

Basic idea

A source may support a proposition without specifying exactly which element of the proposition is true.

For an event A :

- $\text{Bel}(A)$: degree to which evidence supports A ;
- $\text{Pl}(A)$: degree to which evidence is compatible A .

Typically,

$$\text{Bel}(A) \leq \text{Pl}(A).$$

Fuzzy sets and possibility theory

- Possibility theory allows to account imprecise, vague information such as conveyed by natural language.
- Examples: “John is tall”, “the temperature is high”, etc.
- Such information can be represented by a **fuzzy set**, which captures the notion of a set with unsharp boundary. Mathematically, it is a function

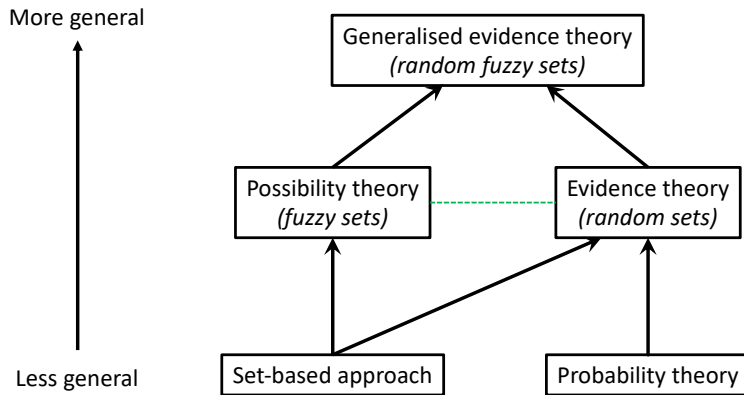
$$\tilde{F} : \Theta \rightarrow [0, 1].$$

- Evidence of the “ X is $\tilde{F}$ ” can be represented by a **possibility measure** $\Pi : \mathcal{P}(\Theta) \rightarrow [0, 1]$ such that

$$\Pi(A) = \sup_{\theta \in A} \tilde{F}(\theta).$$

This is not a probability measure!

A unified view



Random fuzzy sets provide a common framework for sets, probabilities, evidence theory and possibility theory.

Generalised evidence theory

A unified framework based on **random fuzzy sets**.

Guiding idea

Express different forms of uncertainty:

- randomness, as in random sets and probability theory;
- gradual membership, as in fuzzy sets and possibility theory.

This framework allows us to model, combine and propagate several forms of imperfect information within a single mathematical language.

Themes covered in the course

Assuming from basic knowledge of classical uncertainty models: sets and probabilities, the course will introduce the main concepts progressively:

- 1 Evidence theory and belief functions
- 2 Random sets
- 3 Fuzzy sets and possibility theory
- 4 Random fuzzy sets as a unified framework
- 5 Application to uncertainty propagation
- 6 Application to statistical inference
- 7 Application to machine learning

The objective is understand the main concepts, and how to use them.

Part II

Evidence Theory

Roadmap

1 Representation of evidence

- Mass functions
- Belief, plausibility and commonality
- Inversion and properties

2 Combination of evidence

- Dempster's rule of combination
- Discounting

3 Marginalisation and vacuous extension

- Refinements and coarsenings
- Product frames
- Valuation-based systems

Outline

1 Representation of evidence

- Mass functions
- Belief, plausibility and commonality
- Inversion and properties

2 Combination of evidence

3 Marginalisation and vacuous extension

Definitions

Basic setting

- We consider a question Q , to which there exists **one and only one** correct answer X , assumed to belong to some set Θ .
- We call X an **unknown**, and Θ its **frame of discernment**.

In this chapter, we assume that Θ is finite.

Mass function

In evidence theory, we represent evidence about X by assigning positive masses $m(A)$ to subsets $A \subseteq \Theta$, in such a way that the masses sum to 1.

Mass function

A mass function on Θ is a mapping

$$m : \mathcal{P}(\Theta) \rightarrow [0, 1]$$

satisfying

$$m(\emptyset) = 0, \quad \sum_{A \subseteq \Theta} m(A) = 1.$$

- $m(A)$ is the degree to which the evidence supports A as a whole. This mass is not split between elements of A .
- Sets A such that $m(A) > 0$ are called focal sets.

A small example

- Consider an intelligent vehicle equipped with a sensor and a machine learning algorithm recognising objects in the road scene.
- Let X be the class of the object, in the $\Theta = \{\theta_1, \theta_2, \theta_3\}$, with

$$\theta_1 \equiv \text{pedestrian}, \quad \theta_2 \equiv \text{vehicle}, \quad \theta_3 \equiv \text{anything else}.$$

- Assume that the object has been classified as a pedestrian or a vehicle, but there is a 10% probability of sensor malfunction.
- This evidence can be represented by the following mass function:

$$m(\{\theta_1, \theta_2\}) = 0.9, \quad m(\Theta) = 0.1,$$

and $m(A) = 0$ for any $A \in \mathcal{P}(\Theta) \setminus \{\{\theta_1, \theta_2\}, \Theta\}$.

Important distinction

0.90 assigned to $\{\theta_1, \theta_2\}$ is not the same as 0.45 on each singleton.

Special mass functions

A mass function may have up to $2^{|\Theta|} - 1$ focal sets. However, in practice, we often define mass functions with a simple form (as in the previous example). A mass function is

Simple if for some $A \subset \Theta$,

$$m(A) = s, \quad m(\Theta) = 1 - s.$$

Categorical if $m(A) = 1$ for one set $A \subset \Theta$; it represents certainty that the truth belongs to A .

Vacuous if $m(\Theta) = 1$; it represents complete ignorance.

Consonant if focal sets are nested: $A_1 \subseteq A_2 \subseteq \dots \subseteq A_k$.

Bayesian if all focal sets are singletons; then m is equivalent to an ordinary probability mass function.

Random-set interpretation

- Let $(\Omega, \mathcal{P}(\Omega), P)$ be a finite probability space and let $\overline{X} : \Omega \rightarrow \mathcal{P}(\Theta)$. The tuple

$$(\Omega, \mathcal{P}(\Omega), P, \Theta, \mathcal{P}(\Theta), \overline{X})$$

is a (finite) **random set**.

- A random set induces a mass function

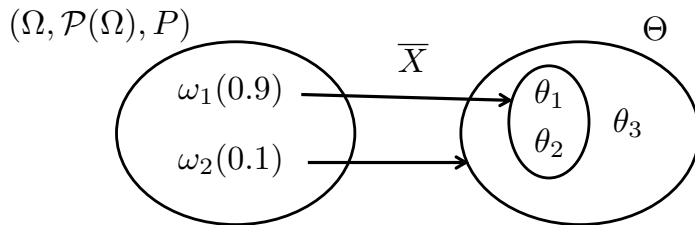
$$m(A) = P\{\omega : \overline{X}(\omega) = A\}.$$

- Meaning:

- ▶ Ω : set of **interpretations** of the evidence;
- ▶ If $\omega \in \Omega$ holds, we know that $X \in \overline{X}(\omega)$, and nothing more.

Back to the example

Here, $\Omega = \{\omega_1, \omega_2\}$, where ω_1 stands for “the sensor is working”, and ω_2 stands for “the sensor is faulty”.



Three transforms of a mass function

For any $A \subseteq \Theta$,

$$\text{Bel}(A) = \sum_{B \subseteq A} m(B), \quad \text{Pl}(A) = \sum_{B \cap A \neq \emptyset} m(B), \quad Q(A) = \sum_{B \supseteq A} m(B).$$

Belief degree of support for A .

Plausibility degree of consistency with A (lack of support for A^c).

Commonality degree of support for *all* elements of A jointly.

$$\text{Bel}(A) \leq \text{Pl}(A), \quad \text{Pl}(A) = 1 - \text{Bel}(A^c).$$

Two-dimensional representation

The uncertainty about A is represented by **two numbers**: $\text{Bel}(A)$ and $\text{Pl}(A)$, with $\text{Bel}(A) \leq \text{Pl}(A)$.

Special cases

- m vacuous: $\text{Bel}(A) = 0$ for all $A \neq \Theta$, and $\text{Pl}(A) = 1$ for all $A \neq \emptyset$.
- m Bayesian: $\text{Bel}(A) = \text{Pl}(A)$ for all A , and Bel is a probability measure.

$$\text{Bel}(A) + \text{Bel}(A^c) \leq 1, \quad \text{Pl}(A) + \text{Pl}(A^c) \geq 1.$$

Computing the transforms in the example

A	$\emptyset$	$\{\theta_1\}$	$\{\theta_2\}$	$\{\theta_1, \theta_2\}$	$\{\theta_3\}$	$\{\theta_1, \theta_3\}$	$\{\theta_2, \theta_3\}$	Θ
$m(A)$	0	0	0	0.90	0	0	0	0.10
$\text{Bel}(A)$	0	0	0	0.90	0	0	0	1
$\text{Pl}(A)$	0	1	1	1	0.10	1	1	1
$Q(A)$	1	1	1	1	0.10	0.10	0.10	0.10

Möbius inversion

The transformations are invertible.

From belief to mass

$$m(A) = \sum_{B \subseteq A} (-1)^{|A|-|B|} \text{Bel}(B).$$

From commonality to mass

$$m(A) = \sum_{B \supseteq A} (-1)^{|B|-|A|} Q(B).$$

Therefore, m , Bel , Pl and Q are **equivalent representations** of the same evidence.

Key boundary conditions

We have

$$\text{Bel}(\emptyset) = \text{Pl}(\emptyset) = 0,$$

$$\text{Bel}(\Theta) = \text{Pl}(\Theta) = 1,$$

but

$$Q(\emptyset) = 1, \quad Q(\Theta) = m(\Theta).$$

- Bel and Pl are monotone:

$$A \subseteq B \Rightarrow \begin{cases} \text{Bel}(A) \leq \text{Bel}(B) \\ \text{Pl}(A) \leq \text{Pl}(B) \end{cases}.$$

- Q is antitone: $A \subseteq B \Rightarrow Q(A) \geq Q(B)$.

Belief and plausibility are not additive

- For any subsets A and B ,

$$\text{Bel}(A \cup B) \geq \text{Bel}(A) + \text{Bel}(B) - \text{Bel}(A \cap B),$$

where the inequality may be strict. We say that belief functions are **super-additive**.

- Conversely,

$$\text{PI}(A \cup B) \leq \text{PI}(A) + \text{PI}(B) - \text{PI}(A \cap B),$$

We say that plausibility functions are **subadditive**.

Reason

Masses assigned to sets intersecting both A and B support $A \cup B$ without supporting either A or B separately.

Plausibility of singletons: contour function

- The **contour function** is

$$\text{pl}(\theta) = \text{Pl}(\{\theta\}) = \sum_{A \ni \theta} m(A).$$

- It measures how plausible each singleton is.
- It does not generally determine the whole mass function, except in special cases.

Consonant case

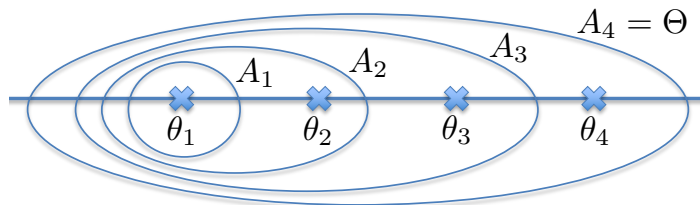
If m is consonant, then

$$\text{Pl}(A) = \max_{\theta \in A} \text{pl}(\theta).$$

For all A, B , we then have

$$\text{Pl}(A \cup B) = \max\{\text{Pl}(A), \text{Pl}(B)\}, \quad \text{Bel}(A \cap B) = \min\{\text{Bel}(A), \text{Bel}(B)\}$$

From the contour function to the mass function



- Let pl be a contour function on the frame $\Theta = \{\theta_1, \dots, \theta_n\}$, with elements arranged by decreasing order of plausibility, i.e.,

$$1 = pl(\theta_1) \geq pl(\theta_2) \geq \dots \geq pl(\theta_n),$$

and let A_i denote the set $\{\theta_1, \dots, \theta_i\}$, for $1 \leq i \leq n$.

- The corresponding mass function is

$$m(A_i) = pl(\theta_i) - pl(\theta_{i+1}), \quad 1 \leq i \leq n-1,$$

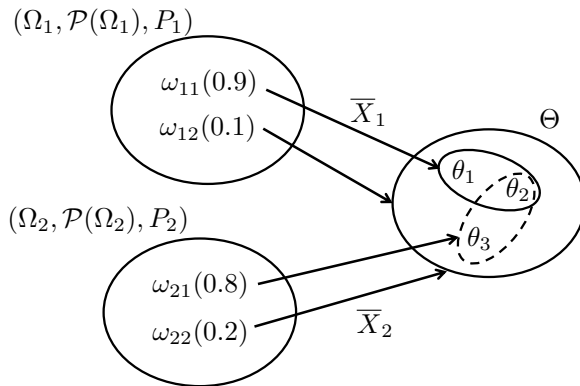
$$m(\Theta) = pl(\theta_n).$$

Outline

- 1 Representation of evidence
- 2 Combination of evidence
 - Dempster's rule of combination
 - Discounting
- 3 Marginalisation and vacuous extension

Combination of evidence (no conflict)

A second sensor tells us that the object is not a pedestrian, and this sensor has 80% chance of functioning correctly.



Combined random set and mass function

- Under this independence assumption, we have the joint probability measure:

$$(P_1 \otimes P_2)(\{\omega_{11}, \omega_{21}\}) = 0.72, \quad (P_1 \otimes P_2)(\{\omega_{11}, \omega_{22}\}) = 0.18,$$

$$(P_1 \otimes P_2)(\{\omega_{12}, \omega_{21}\}) = 0.08, \quad (P_1 \otimes P_2)(\{\omega_{12}, \omega_{22}\}) = 0.02,$$

- With the set-valued function

$$\bar{X}_1 \oplus \bar{X}_2 : (\omega_1, \omega_2) \mapsto \bar{X}_1(\omega_1) \cap \bar{X}_2(\omega_2),$$

we get yields the following combined mass function

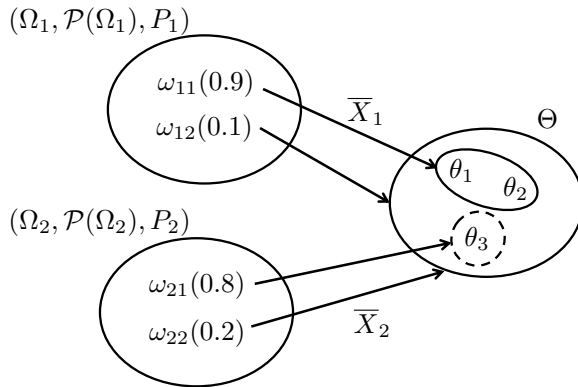
$$(m_1 \oplus m_2)(\{\theta_2\}) = 0.72, \quad (m_1 \oplus m_2)(\{\theta_1, \theta_2\}) = 0.18,$$

$$(m_1 \oplus m_2)(\{\theta_2, \theta_3\}) = 0.08 \quad (m_1 \oplus m_2)(\Theta) = 0.02.$$

- Mass function $m_1 \oplus m_2$ is called the **orthogonal sum** of m_1 and m_2 .

Conflict and normalisation

Assume now that the second piece of evidence tells us that the object is neither a pedestrian, nor a vehicle.



Combined random set and mass function

- Assuming that both sensors are working leads to a contradiction. We must, therefore, conclude that at least one of the two sensors must be faulty, and condition the product probability measure $P_1 \otimes P_2$ on the set of possible pairs in $\Omega_1 \times \Omega_2$.
- Denoting the conditional probability measure as

$$P_{12}^* = (P_1 \otimes P_2)(\cdot \mid \{(\omega_{11}, \omega_{21})\}^c),$$

we get

$$P_{12}^*(\{\omega_{11}, \omega_{22}\}) = 0.18/0.28 = 9/14, \quad P_{12}^*(\{\omega_{12}, \omega_{21}\}) = 0.08/0.28 = 4/14,$$

$$P_{12}^*(\{\omega_{12}, \omega_{22}\}) = 1/14,$$

and the following combined mass function:

$$(m_1 \oplus m_2)(\{\theta_1, \theta_2\}) = 9/14, \quad (m_1 \oplus m_2)(\{\theta_3\}) = 4/14,$$

$$(m_1 \oplus m_2)(\Theta) = 1/14.$$

Dempster's rule

Let m_1 and m_2 be two mass functions on the same frame Θ .

Degree of conflict

$$\kappa = \sum_{B \cap C = \emptyset} m_1(B)m_2(C)$$

Orthogonal sum

If $\kappa < 1$, we define the **orthogonal sum** of m_1 and m_2 as

$$(m_1 \oplus m_2)(A) = \frac{\sum_{B \cap C = A} m_1(B)m_2(C)}{1 - \kappa}, \quad A \neq \emptyset,$$

with $(m_1 \oplus m_2)(\emptyset) = 0$.

Another example

A	$\emptyset$	$\{a\}$	$\{b\}$	$\{a, b\}$	$\{c\}$	$\{a, c\}$	$\{b, c\}$	$\{a, b, c\}$
$m_1(A)$	0	0	0.5	0.2	0	0.3	0	0
$m_2(A)$	0	0.1	0	0.4	0.5	0	0	0

		m_2		
		$\{a\}, 0.1$	$\{a, b\}, 0.4$	$\{c\}, 0.5$
m_1	$\{b\}, 0.5$	$\emptyset, 0.05$	$\{b\}, 0.2$	$\emptyset, 0.25$
	$\{a, b\}, 0.2$	$\{a\}, 0.02$	$\{a, b\}, 0.08$	$\emptyset, 0.1$
	$\{a, c\}, 0.3$	$\{a\}, 0.03$	$\{a\}, 0.12$	$\{c\}, 0.15$

The degree of conflict is $\kappa = 0.05 + 0.25 + 0.1 = 0.4$. The combined mass function is

$$(m_1 \oplus m_2)(\{a\}) = (0.02 + 0.03 + 0.12)/0.6 = 0.17/0.6$$

$$(m_1 \oplus m_2)(\{b\}) = 0.2/0.6$$

$$(m_1 \oplus m_2)(\{a, b\}) = 0.08/0.6$$

$$(m_1 \oplus m_2)(\{c\}) = 0.15/0.6.$$

Commonality under combination

The commonality transform makes conjunctive combination simple:

$$Q_{m_1 \oplus m_2}(A) = \begin{cases} \frac{Q_{m_1}(A)Q_{m_2}(A)}{1-\kappa}, & A \neq \emptyset, \\ 1 & A = \emptyset. \end{cases}$$

Practical message

Dempster's rule is multiplicative in the commonality domain.

Algebraic properties

For normal mass functions on the same frame, when the rule is defined:

- **Commutativity:** $m_1 \oplus m_2 = m_2 \oplus m_1$.
- **Associativity:** $(m_1 \oplus m_2) \oplus m_3 = m_1 \oplus (m_2 \oplus m_3)$.
- **Neutral element:** the vacuous mass $m_\Theta(\Theta) = 1$.

Consequence

Several independent pieces of evidence can be combined in any order, and ignorance changes nothing.

What Dempster's rule assumes

Dempster's rule is appropriate when

- sources are independent and reliable;
- the frame is correctly specified.

Warning

High conflict may indicate unreliable sources, dependence, or a wrong frame. In some cases, the initial model assumptions must be revised.

Conditioning as a special case

- **Conditioning** a mass function m on $B \subseteq \Theta$ is combination with the categorical mass $m_B(B) = 1$:

$$m(\cdot | B) = m \oplus m_B.$$

- Thus, for $A \neq \emptyset$,

$$m(A | B) = \frac{\sum_{C \cap B = A} m(C)}{\text{Pl}(B)}.$$

It is well defined iff $\text{Pl}(B) > 0$.

Property

$$\text{Pl}(A | B) = \frac{\text{Pl}(A \cap B)}{\text{Pl}(B)}.$$

Generalises Bayesian conditioning.

Computational cost

A direct combination over $\mathcal{P}(\Theta)$ can be expensive.

- Number of subsets: $2^{|\Theta|}$.
- Pairwise intersections of focal sets: $|\text{Foc}(m_1)| |\text{Foc}(m_2)|$.
- Repeated fusion may generate many focal sets.

Key idea

Exploit structure: sparse focal sets, product frames, graphical decompositions.

Meta-evidence about a source

A source may be unreliable.

- A sensor may fail.
- An expert may be only partially trustworthy.
- A classifier may be trained outside its domain of validity.

Let $\alpha \in [0, 1]$ denote the reliability of the source.

Principle

Keep a fraction α of the source's committed evidence and transfer the remaining mass to total ignorance.

Discounting operation

For a mass function m and reliability α ,

$$\begin{aligned}\alpha m(A) &= \alpha m(A), & A \subset \Theta, \\ \alpha m(\Theta) &= 1 - \alpha + \alpha m(\Theta).\end{aligned}$$

The quantity $1 - \alpha$ is called the **discount rate**.

Special cases

- $\alpha = 1$: no discounting.
- $\alpha = 0$: vacuous mass function.
- Lower reliability means less commitment and wider belief–plausibility intervals.

Effect on belief and plausibility

For $A \neq \Theta$,

$$\text{Bel}_{\alpha m}(A) = \alpha \text{Bel}_m(A),$$

whereas

$$\text{Pl}_{\alpha m}(A) = 1 - \alpha + \alpha \text{Pl}_m(A) \quad (A \neq \emptyset).$$

Interpretation

Discounting decreases belief in proper events and increases plausibility of their complements: it expresses loss of confidence.

Discounting before combination

When combining several sources, reliability should normally be taken into account before fusion:

$$^{\alpha}m_1 \oplus ^{\beta}m_2.$$

- Discounting reduces the influence of unreliable sources.
- It may also reduce conflict.
- Reliability is **meta-evidence** about the source, not evidence about the state itself.

Outline

- 1 Representation of evidence
- 2 Combination of evidence
- 3 Marginalisation and vacuous extension**
 - Refinements and coarsenings
 - Product frames
 - Valuation-based systems

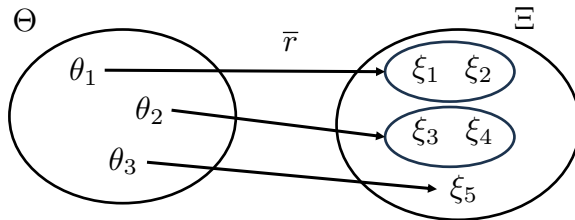
Why change the frame?

- The definition of a frame is often a matter of convention.
- In the object recognition example, we could have defined a finer frame distinguishing, e.g.,
 - ▶ different kinds of vehicles (bicycles, cars, etc.);
 - ▶ different kinds of objects in the “anything else” class (such as “animal”, “road sign”, etc.).
- Different sources of information may provide evidence represented in frames of different granularities.

Goal

Express the same piece of evidence with different levels of detail without adding unwarranted information.

Refinement and coarsening



- A frame Ξ is a **refinement** of Θ iff there exists a mapping (called a **refining**)

$$\bar{r} : \mathcal{P}(\Theta) \rightarrow \mathcal{P}(\Xi)$$

such that

- ▶ the subsets $\{\bar{r}(\{\theta\}) : \theta \in \Theta\}$ form a partition of Ξ ;
- ▶ for all $A \subseteq \Theta$, $\bar{r}(A) = \bigcup_{\theta \in A} \bar{r}(\{\theta\})$.
- We then say that Θ is a **coarsening** of Ξ .

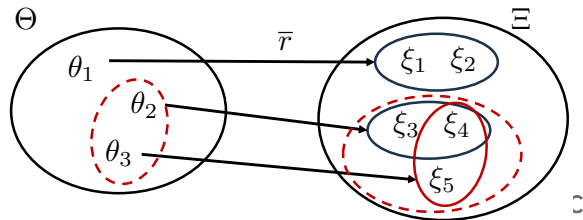
Vacuous extension and restriction

- From Θ to Ξ : given a mass function m^Θ on Θ , its **vacuous extension** $m^{\Theta \uparrow \Xi}$ in Ξ is obtained by transferring each mass $m^\Theta(A)$ to $\bar{r}(A)$.
- From Ξ to Θ : given a mass function m^Ξ on Ξ , its **restriction** $m^{\Xi \downarrow \Theta}$ in Θ is obtained by transferring each mass $m^\Xi(B)$ to its **upper inverse**

$$B^* := \{\theta \in \Theta : \bar{r}(\{\theta\}) \cap B \neq \emptyset\}.$$

- Example: $\{\xi_4, \xi_5\}^* = \{\theta_1, \theta_2\}$.

Going from a refined frame to a coarser frame loses detail.



Product frames

- Let $X_1, \dots, X_n$ be n unknowns with frames $\Theta_1, \dots, \Theta_n$.
- The product space is

$$\Theta = \Theta_1 \times \dots \times \Theta_n.$$

- For $I \subseteq \{1, \dots, n\}$, write

$$\Theta_I = \prod_{i \in I} \Theta_i,$$

and

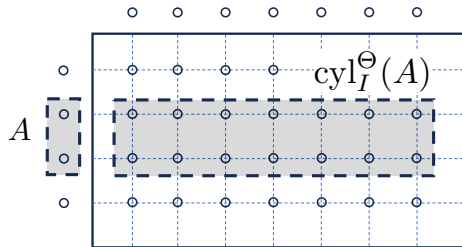
$$\Theta = \Theta_I \times \Theta_J, \quad J = I^c.$$

Vacuous extension

The **cylindrical extension** of $A \subseteq \Theta_I$ in Θ is

$$\text{cyl}_I^\Theta(A) = A \times \Theta_J.$$

The mapping $\text{cyl}_I^\Theta : A \mapsto \text{cyl}_I^\Theta(A)$ is a refining.



Vacuous extension

Let m^{Θ_I} be a mass function on Θ_I . Its vacuous extension to Θ is

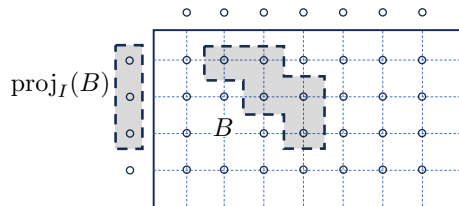
$$m^{\Theta_I \uparrow \Theta}(B) = \begin{cases} m^{\Theta_I}(A) & \text{if } B = \text{cyl}_I^\Theta(A) \text{ for some } A \subseteq \Theta_I, \\ 0, & \text{otherwise.} \end{cases}$$

Each mass $m^{\Theta_I}(A)$ is transferred to the cylindrical extension of A .

Marginalisation

The **upper inverse** of $B \subseteq \Theta$ in Θ is its **projection**, defined as

$$\begin{aligned}\text{proj}_I(B) &= \{x_I \in \Theta_I : \text{cyl}_I^\Theta(\{x_I\}) \cap B \neq \emptyset\} \\ &= \{x_I \in \Theta_I : \exists x_J \in \Theta_J, (x_I, x_J) \in B\}.\end{aligned}$$



Marginal

Let m be a mass function on Θ . Its **marginal** on Θ_I is

$$m^{\Theta \downarrow \Theta_I}(A) = \sum_{B \subseteq \Theta: \text{proj}_I(B)=A} m^\Theta(B).$$

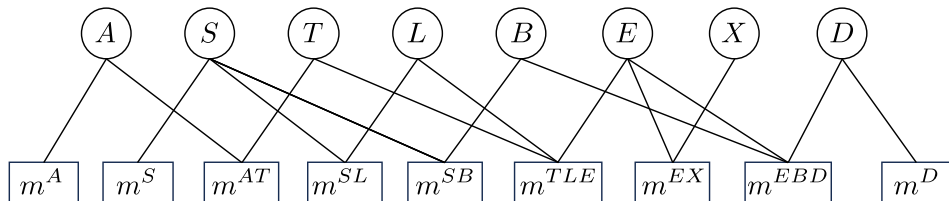
- Each joint focal set is projected onto the variables of interest.
- Marginalisation discards information about the remaining variables.
- It is the belief-function analogue of probabilistic marginalisation.

Valuation-based systems

- Many problems can be formalised by defining
 - ▶ n unknown $X_1, \dots, X_n$ in frames Θ_i , $i = 1, \dots, n$, and
 - ▶ q mass functions $m_1, \dots, m_q$, each m_i being defined in frame $\Theta_{I(i)} = \prod_{i=1}^{I(i)} \Theta_i$ for some subset $I(i) \subseteq \{1, \dots, n\}$.
- Such a model is called an **evidential valuation-based system (VBS)**.

Chest clinic example

*Shortness-of-breath (Dyspnoea) may be due to **T**uberculosis, **L**ung cancer or **B**ronchitis, or none of them, or more than one of them. A recent visit to **A**sia increases the chances of tuberculosis, while **S**moking is known to be a risk factor for both lung cancer and bronchitis. The results of a single chest **X**-ray detects that the patient has **E**ither lung cancer and tuberculosis; dyspnoea also does not discriminate between lung cancer and tuberculosis.*



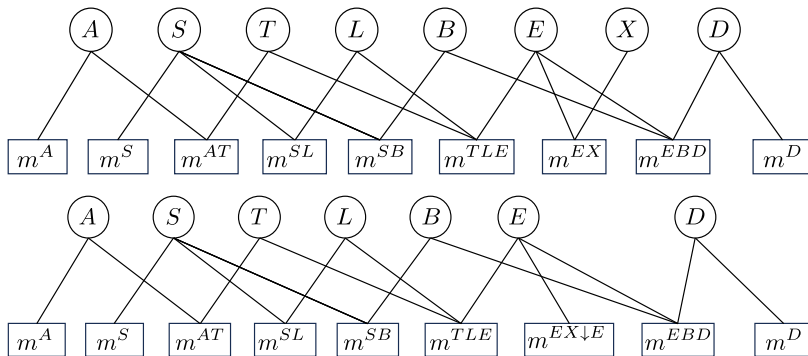
Fusion algorithm: local combination and deletion

- Denote by X_J , $J \subset \{1, \dots, n\}$ the subset of **target variables**.
- We want to compute

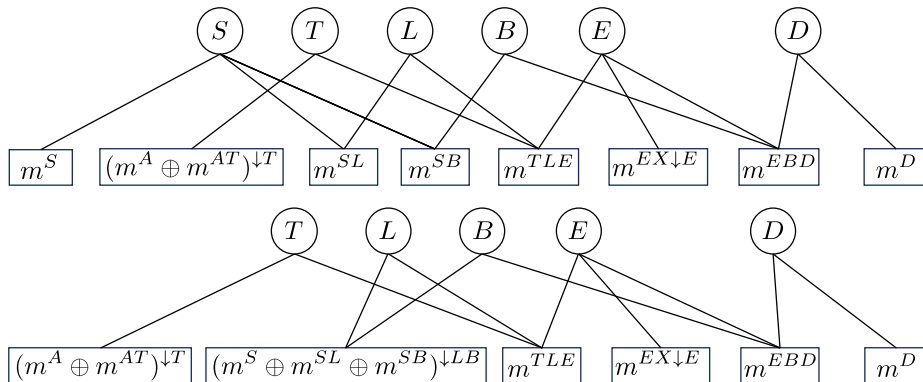
$$m^{\Theta_J} = (m_1 \oplus \dots \oplus m_q)^{\downarrow \Theta_J}.$$

- The **Fusion algorithm** deletes variables in J^c one at a time by iterating the following steps:
 - ① combine all valuations involving a variable to be eliminated;
 - ② marginalise the result to the remaining variables;
 - ③ continue until only target variables remain.
- Localisation reduces complexity when each piece of evidence concerns few variables.
- Different deletion sequences may induce different computational costs.
- **One-step lookahead** heuristic: select as the next unknown for deletion the one that leads to combination over the smallest state space, with ties broken randomly.

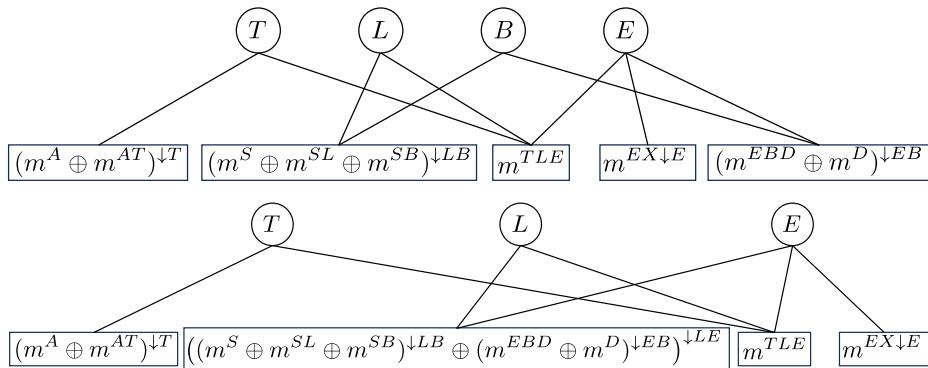
Example (1/4)



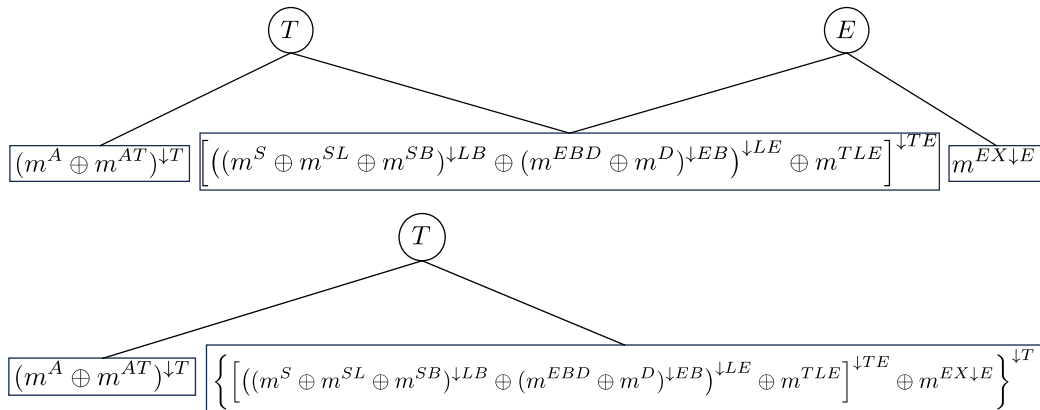
Example (2/4)



Example (3/4)



Example (4/4)



Summary of the lecture

- A mass function allocates evidence to subsets of the frame.
- Belief, plausibility and commonality are equivalent transforms with different interpretations.
- Dempster's rule combines independent evidence by intersecting random sets and conditioning on non-empty intersections.
- Discounting expresses partial reliability of sources.
- Vacuous extension and marginalisation allow evidence to be handled in product frames.
- The Fusion algorithm uses the local structure of the evidence to combine and marginalise mass functions efficiently.

Further reading

cf. <http://www.hds.utc.fr/~tdenoeux>



G. Shafer.

A Mathematical Theory of Evidence.

Princeton University Press, 1976.



T. Denœux, D. Dubois and H. Prade.

Representations of Uncertainty in Artificial Intelligence: Beyond Probability and Possibility.

In P. Marquis, O. Papini and H. Prade (Eds), "A Guided Tour of Artificial Intelligence Research", Volume 1, Chapter 4, Springer Verlag, pages 119-150, 2020.



T. Denœux.

Modern Evidence Theory: A Unified Approach to Uncertainty in AI, Engineering and Statistics.

Book manuscript (forthcoming).